

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 9: Consolidating the Evidence and Building the Final Model: Why Did Mr. Wafula's Maize and Beans Behave So Differently?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
D	D	C	C	C	A	A	C

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on the three puzzles and 'If I changed only the rain, would the differences disappear?'

Q1. Mr. Wafula planted maize and beans on the same day, in the same field, after the same 25 mm of rain. Yet the two crops differed in germination, emergence and growth. Which statement best explains the differences?

- A. The two crops were planted on different days.
- B. The beans received less rain than the maize.
- C. Healthy-looking seeds always germinate, so the differences were only due to chance.
- D. The differences come from seed structure, seed condition (such as dormancy or viability) and the environment (such as waterlogged patches).**

Correct answer: D. Rain and planting date were the same for both crops, so they cannot explain the differences. The lesson shows that seed structure, seed condition and environmental factors such as waterlogging do.

USE AFTER: OBSERVE PHASE — Observe: Card B, scarification of the 8 hard-coated beans (7 of 8 germinated by Day 7)

Q2. Out of 100 Rosecoco bean seeds, 20 did not germinate by Day 14. Of these, 8 had a hard seed coat. When the hard coats were nicked (scarified), 7 of the 8 seeds germinated. What does this show about the 8 hard-coated seeds?

- A. They needed more rain because they were planted too deep.
- B. They were eaten by pests, and nicking removed the pests.
- C. They were dead (non-viable) and germinated by chance.
- D. They were alive but dormant, because the hard seed coat stopped water from entering.**

Correct answer: D. Seeds that germinate after the coat is nicked were viable. The hard, impermeable testa kept out water, so dormancy is not the same as death.

USE AFTER: OBSERVE PHASE — Observe: Card C, jar results for 20 bean seeds at Day 5

Q3. Twenty bean seeds were placed in each of four jars for 5 days. Results: water + air + 25 °C = 18 germinated; dry seeds = 0 germinated; water but no air = 1 germinated; water + air at 4 °C = 0 germinated. What is the best conclusion?

- A. Only water is needed for germination.
- B. Light is needed for germination, because the seeds in the warm jar germinated.
- C. Seeds need water, oxygen and a suitable temperature together to germinate.**
- D. Cold temperatures speed up germination.

Correct answer: C. Removing water, air or warmth each reduced germination to 0 or 1, while the jar with all three gave 18 of 20. All three conditions are needed.

USE AFTER: EXPLAIN PHASE — Explain: CER for puzzle 3, maize growth steepest between Day 21 and Day 35 (Card E heights)

Q4. The height of a maize plant was 62 cm on Day 28 and 98 cm on Day 35. By how many centimetres did it grow during that week?

- A. 27 cm
- B. 98 cm
- C. 36 cm**
- D. 32 cm

Correct answer: C. Increase in height = $98\text{ cm} - 62\text{ cm} = 36\text{ cm}$. This is the fastest weekly increase, showing the steep middle part of the sigmoid growth curve.

USE AFTER: EXPLAIN PHASE — Explain: CER for puzzle 3, stem width (Card E) and maize having no true secondary growth

Q5. A maize stem became thicker, from 6 mm at Day 14 to 22 mm at Day 42, even though maize has no vascular cambium. How does the maize stem thicken?

- A. Thickening is not growth, because growth means only an increase in height.
- B. It does not really grow thicker; the stem only swells with water.
- C. By primary growth, through cell division and enlargement of tissues produced at the apical meristem.**
- D. By secondary growth from the vascular cambium, as in the bean.

Correct answer: C. Maize has no vascular cambium, so true secondary growth does not occur. Its thickening comes from primary growth, while growth means a permanent, irreversible increase in size, not only in height.

USE AFTER: EXPLAIN PHASE — Explain: CER for puzzle 2, linking hypogeal and epigeal patterns to seed structure

Q6. A bean seedling emerges with its seed leaves (cotyledons) lifted above the soil. A maize seedling emerges with its scutellum staying underground. Which description is correct?

- A. Bean emergence is epigeal and maize emergence is hypogeal.**
- B. Both bean and maize emergence are hypogeal.
- C. Both bean and maize emergence are epigeal.
- D. Bean emergence is hypogeal and maize emergence is epigeal.

Correct answer: A. Epigeal emergence lifts the cotyledons above the soil (bean). Hypogeal emergence leaves the seed storage tissue underground (maize, where the scutellum stays in the soil).

USE AFTER: EXPLAIN PHASE — Explain: Card F hormone evidence, apical dominance (intact tip 0 branches, tip removed 3 branches)

Q7. A bean plant with its tip intact had 0 branches at Day 25. Another bean plant had its tip removed at Day 20 and had 3 branches by Day 32. What explains the branching?

- A. Auxin from the apical bud inhibits the lateral buds (apical dominance); removing the tip removes this effect, and cytokinins help the lateral buds grow.**
- B. The tip makes abscisic acid, which makes the plant grow lateral branches when removed.
- C. Auxin from the apical bud promotes lateral buds, so removing the tip lowers branching.
- D. Gibberellin from the tip stops branches from forming, and cutting adds more gibberellin.

Correct answer: A. Apical dominance means auxin from the apical bud inhibits lateral buds. Removing the tip lets cytokinins promote them, giving 3 branches instead of 0.

USE AFTER: MODEL BUILDING PHASE — Model: final model section on causes of failure by crop (maize 5, 4, 3 and beans 8, 6, 4, 2) and the waterlogged patch

Q8. In Mr. Wafula's field, 3 maize seeds and 6 bean seeds failed to germinate because they were in a waterlogged patch. Why does waterlogging stop germination?

- A. Waterlogged soil makes the seed coat hard, which causes dormancy.
- B. Waterlogged soil is always colder than 4 °C.
- C. The water fills the air spaces in the soil, so the seeds get too little oxygen for respiration.**
- D. Too much water makes the seed produce too much gibberellin.

Correct answer: C. Waterlogging fills soil air spaces with water, which limits oxygen. Germinating seeds need oxygen for respiration, so waterlogged seeds fail even though they are viable.